Knowledge-Distilled TinyML Bird/Non-Bird Acoustic Classifier

Two-Page Executive Summary for Supervisor Review

Researcher: Ali Amini **Target platform:** Seeed Studio XIAO ESP32S3

Objective and scope

Develop a compact binary classifier that processes a fixed five-second audio window directly on an ESP32S3-class edge device:

5 s audio 16 kHz mono → 40 × 249 log-mel → INT8 DS-CNN → p_{bird} decision

Environmental sensors, LoRaWAN, cloud communication, scheduling, and final battery sizing are intentionally outside the present scope. They are deferred until physical live-audio inference is demonstrated.

What has been completed

- Audited the 27-class FSC22 SWinT-BiLSTM Teacher and recovered the correct bird-output mapping: `birdchirping` → 12, `wingflapping` → 20.
- Generated deterministic binary Teacher probabilities and trained compact Student models using hard labels plus knowledge distillation.
- Performed a KD-weight sweep and selected **0.8 hard focal loss + 0.2 soft-BCE KD**.
- Trained the selected Student on the full dataset and exported FP32 and full-integer INT8 TFLite models.
- Reimplemented the full log-mel front-end in C++, verified Python/C++ numerical parity, and replaced the direct DFT with a memory-safe ESP-DSP FFT path.
- Added the embedded model asset, TensorFlow Lite Micro wrapper, direct-input smoke test, preprocessing-to-inference firmware, and optimized ESP32S3 build configuration.
- All relevant ESP32S3 projects currently **compile and link successfully**.

Selected model and deployment contract

Input duration	5.0 s
Sample rate	16,000 Hz
Samples	80,000 mono
Feature tensor	40 × 249 × 1
Architecture	DS-CNN

Parameters	32,097
Normalization	$\mu = -9.19348$, $\sigma = 3.78920$
Decision threshold	0.6734629869
INT8 model size	59,760 bytes
Preprocess scratch	5,124 bytes

Model-selection evidence (cross-validation)

Precision	Recall	F1	PR-AUC	ROC-AUC	Recall @ FPR=0.02
0.728	0.873	0.780	0.905	0.986	0.860

Interpretation: these cross-validation results are the valid model-selection evidence. Metrics obtained after final training on the complete dataset are apparent training metrics and are not presented as independent test performance.

Embedded validation status and exact next step

Verified engineering results

Teacher mapping and deterministic soft labels	Verified
Student baselines, KD sweep, and selected configuration	Verified
FP32/INT8 TFLite export and Keras agreement	Verified
Initial Python/C++ preprocessing parity: 99.9598% exact, max error 1 INT8 unit	Verified
Optimized FFT parity: 100% exact, mean INT8 difference 0	Verified
Memory-safe frame-by-frame preprocessing with ESP-DSP	Verified
TFLite Micro wrapper and ESP32S3 compile/link tests	Verified

What has not yet been demonstrated

Five-second waveform capture on the physical board	Pending
Physical end-to-end inference on live audio	Pending
Measured preprocessing and inference latency	Pending
Measured SRAM/PSRAM peaks and runtime stability	Pending
Controlled bird/non-bird playback evaluation	Pending
Measured energy per classification and final battery capacity	Pending

Current blocker

The available **Grove Sound Sensor v1.7** is an analog sound-intensity detector. The manufacturer explicitly states that it should not be used as a recording device. It can later be used as a sound/no-sound trigger, but it cannot provide the five-second waveform required by the classifier.

The prepared live-audio firmware targets the digital PDM microphone on the **XIAO ESP32S3 Sense expansion board** using:

```
PDM DATA = GPIO41     PDM CLK = GPIO42
```

That microphone board is not currently available; therefore, physical live-audio validation is blocked by hardware rather than by the trained model or embedded software.

Required purchase

Preferred: purchase a **Seeed Studio XIAO ESP32S3 Sense kit** and confirm that it includes the Sense expansion board with the digital microphone. This path matches the PDM firmware already prepared.

Alternative: purchase a digital I2S microphone breakout such as INMP441, ICS-43434, or SPH0645. This is technically valid but requires a new standard-I2S capture backend and explicit wiring.

Exact next validation after the microphone arrives

1. Capture exactly 80,000 signed 16-bit samples at 16 kHz.
2. Verify duration, sample count, RMS, peak, DC offset, zeros, and clipping.
3. Execute the optimized ESP-DSP log-mel preprocessing.
4. Run the INT8 TFLite Micro Student and report p_{bird} and the decision.
5. Record preprocessing latency, inference latency, total latency, internal RAM, and PSRAM.
6. Test controlled bird and non-bird playback, then measure current during capture, preprocessing, inference, and sleep.

Decision requested from supervisor: approve procurement of the waveform-capable digital microphone hardware and continuation with the physical five-second end-to-end validation. No work on environmental sensors or LoRaWAN is required before this milestone passes.

Repository checkpoint: `2f3921e` – project status and hardware blocker documented.

Supporting documentation: full technical report, final Student model card, preprocessing parity reports, and ESP32S3 firmware projects are available in the GitHub repository.

Hardware references: `https://wiki.seeedstudio.com/xiao_esp32s3_sense_mic/`; `https://wiki.seeedstudio.com/Grove-Sound_Sensor/`